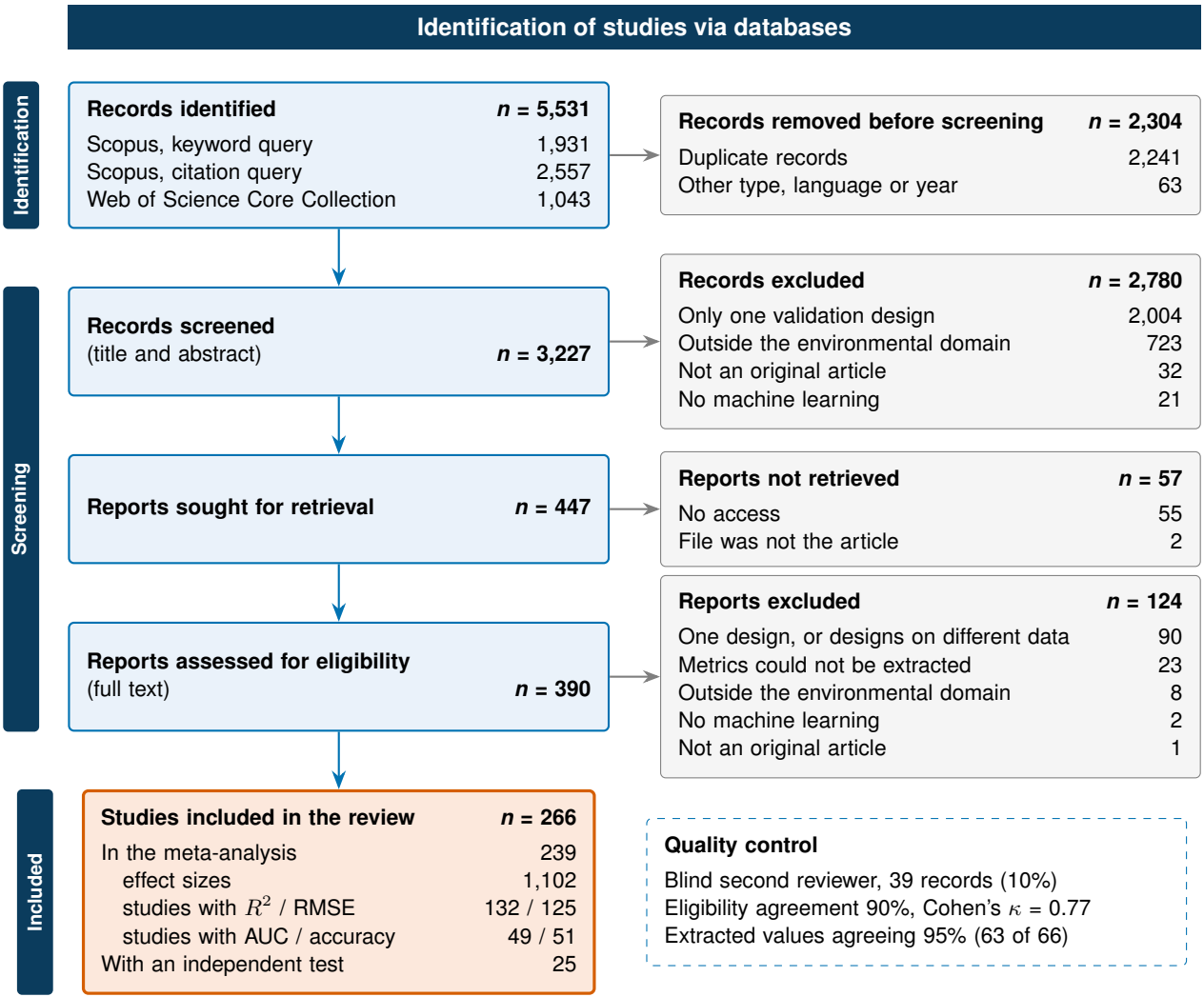
Identification of studies via databases
Identification
Screening
Included
Records identified n = 5,531
Scopus, keyword query 1,931
Scopus, citation query 2,557
Web of Science Core Collection 1,043
Records removed before screening n = 2,304
Duplicate records 2,241
Other type, language or year 63
Records screened
(title and abstract) n = 3,227
Records excluded n = 2,780
Only one validation design 2,004
Outside the environmental domain 723
Not an original article 32
No machine learning 21
Reports sought for retrieval n = 447
Reports not retrieved n = 57
No access 55
File was not the article 2
Reports assessed for eligibility
(full text) n = 390
Reports excluded n = 124
One design, or designs on different data 90
Metrics could not be extracted 23
Outside the environmental domain 8
No machine learning 2
Not an original article 1
Studies included in the review n = 266
In the meta-analysis 239
effect sizes 1,102
studies with R^2 / RMSE 132 / 125
studies with AUC / accuracy 49 / 51
With an independent test 25
Quality control
Blind second reviewer, 39 records (10%)
Eligibility agreement 90%, Cohen's κ = 0.77
Extracted values agreeing 95% (63 of 66)